

Learning about Version Control

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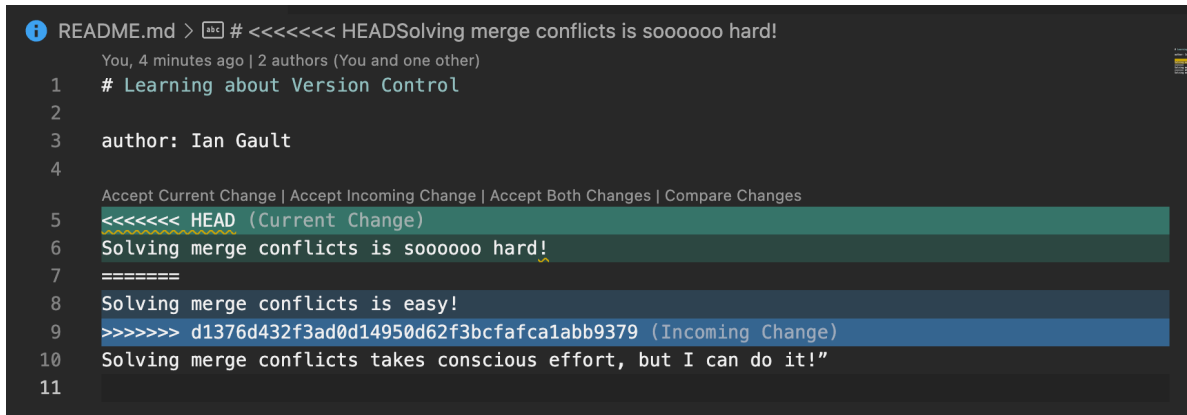
Solving merge conflicts takes conscious effort, but I can do it!

I like version control because it allows for collaboration and quality control of data science projects

A frustration with version control is not remembering all of the commands and what they do.

<https://github.com/iangault/learning-about-version-control/commits/main/>

Branching is useful for being able to split from the main branch and work on a separate branch for development, testing, review. It allows for commits to be made on the main branch, while you are able to version control different parts of the project (parallelization of work) and have checkpoints where isolated changes can become part of the shared history.



The screenshot shows a GitHub merge conflict resolution interface for the file README.md. At the top, it says "You, 4 minutes ago | 2 authors (You and one other)". Below this, the file content is shown with line numbers 1 through 11. Line 1 is "# Learning about Version Control". Line 2 is empty. Line 3 is "author: Ian Gault". Line 4 is empty. Line 5 is the start of a conflict, marked with "<<<<<<< HEAD (Current Change)". Line 6 contains the text "Solving merge conflicts is soooooo hard!". Line 7 is a separator line consisting of six equals signs. Line 8 contains the text "Solving merge conflicts is easy!". Line 9 is the start of another conflict, marked with ">>>>>>> d1376d432f3ad0d14950d62f3bcfafca1abb9379 (Incoming Change)". Line 10 contains the text "Solving merge conflicts takes conscious effort, but I can do it!". Line 11 is empty. Above the conflict lines, there are four buttons: "Accept Current Change", "Accept Incoming Change", "Accept Both Changes", and "Compare Changes".

```
i README.md > # <<<<<<< HEADSolving merge conflicts is soooooo hard!
You, 4 minutes ago | 2 authors (You and one other)
1 # Learning about Version Control
2
3 author: Ian Gault
4
5 <<<<<<< HEAD (Current Change)
6 Solving merge conflicts is soooooo hard!
7 =====
8 Solving merge conflicts is easy!
9 >>>>>>> d1376d432f3ad0d14950d62f3bcfafca1abb9379 (Incoming Change)
10 Solving merge conflicts takes conscious effort, but I can do it!"
11
```

Figure 1: Conflict resolution

```
(522) iangault@dhcp-128-189-192-249 ~/git/MDS/522_workflows/learning-about-  
ersion-control (main *+|MERGING)  
$ git diff  
diff --cc README.md  
index d0733bf,11bb768..0000000  
--- a/README.md  
+++ b/README.md  
@@@ -2,4 -2,4 +2,4 @@@  
  
author: Ian Gault  
  
- Solving merge conflicts is soooooo hard!  
-Solving merge conflicts is easy!  
++Solving merge conflicts takes conscious effort, but I can do it!
```

Figure 2: Git status